

Dylan Ravitch

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EDUCATION

Bachelor of Science, Mechanical Engineering May 2025

Washington State University, Vancouver, WA

- Thermal Design, Heat Transfer, Mechanics of Materials, Manufacturing of Polymer Composites, Machine Design, Sensor Technology, Aerodynamics

Associates of Arts

June 2019

Clark College, Vancouver, WA

PROFESSIONAL EXPERIENCE

Mechanical Engineer 2

May 2024 – Present

SSOE, Hillsboro, OR

- Collaborated with client to structure data and develop constraint-based CSVs enabling automated routing of mechanical and electrical connections.
- Designed fluid delivery systems for semiconductor fabrication facilities, including high-pressure and hazardous gas routing.
- Identified appropriate pipe classes and materials for fluid systems based on chemical compatibility and applicable industry standards.
- Participated in construction design reviews, identifying and resolving constructability issues.

Manufacturing Engineer Intern

May 2021 – January 2023

Pacific Stainless Products, St. Helens, OR

- Created detailed sheet metal part assemblies and fabrication drawings for orders from LAM Research and Intel.
- Coordinated with machinists on the shop floor to troubleshoot fabrication issues and refine designs for manufacturability.

CZ Ingot Growing Operator

November 2023 – May 2024

SEH America, Vancouver, WA

- Operated and troubleshoot high-temperature vacuum furnaces used in silicon crystal growth process.
- Conducted equipment inspections, logged critical data, and maintained system functionality.

PROJECTS

TSMC Washington Chemical Container Capstone

- Designed a 3D-printed polypropylene chemical transport container for TSMC Washington with integrated safety features for hazardous fluid handling.
- Ensured design compliance with OSHA regulations and internal safety standards for hazardous chemical handling.
- Utilized rapid prototyping techniques, including 3D-printed snap-fit features and plastic welding, to iterate and validate functional prototypes.

RC Plane Wing Design Project

- Designed and fabricated a remote-controlled (RC) plane wing using Styrofoam, guided by aerodynamic analysis in XFOIL simulation software.
- Conducted wind tunnel testing to evaluate lift and drag across varying speeds and angles of attack.

Rocketry - Aerospace Club WSUV

- Developed and modeled the main body in collaboration with multiple teams to ensure proper assembly.
- Conducted finite element analysis (FEA) on the airframe using Solidworks.
- Placed 3rd in category and 6th overall at Spaceport America Cup 2022.

SKILLS

Autodesk Inventor, Revit, AutoCAD, Solidworks (CAD & FEA), MATLAB, XFOIL, Visual Basic